

Research article

Match experiences affect interest: Impacts of matchmaking and performance on churn in a competitive game

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ABSTRACT

This study assessed how matchmaking and match results affect player churn in a multiplayer competitive game. In competitive games, matchmaking is crucial in gathering players with similar skills and creating balanced player-versus-player matches. Players are highly motivated when they win matches, whereas losing matches is demotivating, leading to churn. We performed a two-way fixed effects estimation using our panel data to analyze the relationship between players' churn and match experience. The panel data retrieved 42 days of server-side in-game logs, comprising approximately six million matches played by more than 262k players in the casual commercial game "Everybody's Marble." The experimental results indicate that churn is positively influenced by being matched with stronger opponents. Interestingly, being matched with weaker opponents decreases the possibility of churn more than fair matches (being matched with equally skilled opponents). Furthermore, large differences in opponents' skill levels positively influence churn, while more frequent and consecutive wins negatively influence it. The results also reveal that consecutive losses can affect churn differently, depending on the players' level. This study provides theoretical and practical implications for researchers who want to understand the factors that affect user churn and game developers who want to maximize user retention rates in commercial games.

1. Introduction

Managing player churn from launched games is an age-old problem in the game industry. Churn occurs when players leave a game; therefore, the churn rate is a key indicator of whether players are enjoying a game [18]. Acquiring new customers is less cost-effective than retaining current customers because of advertisement fees [45]. Thus, game companies attempt to maintain low churn rates. Analyzing churn helps us understand what stimulates player interest and motivates them. Consequently, the ability to predict potential churners has improved significantly [2]. Churn prediction studies focus on prediction accuracy, so that game operators can find churners in advance and employ customer acquisition strategies. However, studies have not been able to determine the exact reason for churn. Previous studies have proposed key variables in their churn prediction systems, but most explore symptoms before churn (e.g., session duration and frequency) rather than the causes of churn [29].

This study explored causal factors for churn in a massive multiplayer game, particularly focusing on match-related player experience. Although the reasons for quitting a game vary (e.g., a decline in achievement, negative influence of in-game social

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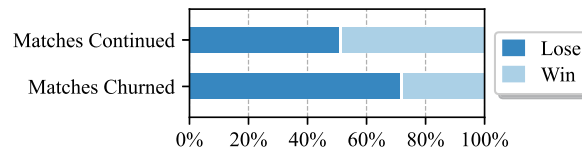


Fig. 1. An example of the relationship between match results (win vs. lose) and churn. *Matches churned* represents the last matches that the users played before leaving the game, while *matches continued* represents the matches where the users continued to play. The observation period was 42 days, consisting of 27 million matches.

activities, or personal preferences and schedules), we specifically examined player performance and competition. Competitive games are widespread across numerous multiplayer online game genres. This immersive content contributes to a better user experience when it offers players the ability to succeed [21]. Unbalanced matches and losing games adversely affect user experiences. Fig. 1 shows a simple example that indicates that losing may cause more churn. The figure compares the proportions of matches where players stayed or left immediately after the match. The percentage of lost matches was higher in cases of churned matches. Similarly, we found other elements that are correlated to the churn rate, such as match opponents or consecutive losses while analyzing our data. Therefore, we formulated the following research questions:

- Do players' match experiences cause churn? Specifically, to what extent do matchmaking results and players' performance directly affect churn?

To answer these questions, we gathered players' skill levels, matchmaking logs, match results, and statuses. (Note that we define skill as the relative strength of a user in a game compared to other users.) To produce statistically reliable results, we analyzed the data of six million matches from 262,698 individuals who played and churned during the observation period. We utilized a two-way fixed-effects model to remove bias caused by the heterogeneity of individuals and time [48]. We constructed panel data by collecting players' daily statuses and applied logistic regression because churn is a binary variable.

Our study contributes to addressing the gap in measuring causal effects on churn. This work enhances the similar work using a simulated game [35], providing results investigated from large-scale actual game data. Our findings show how difficulty and achievements affect churn in player-versus-player matches, in contrast to the studies targeting one-player tasks [39,37,24]. Also, our work focuses more on matchmaking factors, which is distinguished from the studies analyzing other factors related to achievement, social connection, and promotions [5,4]. We found that matching players with highly disproportionate skills caused greater churn. Better performance, characterized by higher win rates or more frequent consecutive wins, made the players churn less. Also, the players' in-game phase was an essential moderator of the churn problem. Performance-related factors became more important to players as their levels rise.

The remainder of this paper is organized as follows: Section 2 reviews the related literature and develops hypotheses. Section 3 describes the research methodology. Section 4 provides empirical analysis results. Section 5 discusses the implications of the findings and concludes the paper.

2. Literature review and hypotheses development

2.1. Player churn

The term *churn* refers to customer turnover or withdrawal from a current service provider [17]. In games, churn means that a user has stopped playing and remained disconnected for a long time. For games with subscriptions, unsubscribed players are churners. However, it is more complicated to define churners in the case of free-to-play games (including the game analyzed in this study) because there is no monetary evidence of game termination. Accordingly, many researchers have considered players who abandon a game for a certain period as churners [18,30,29].

As a game's survival depends on the number of active players, efforts have been made to reduce the number of churners. Research on churn *prediction* is part of these efforts [42]. Researchers have identified various indicators that can help predict potential churners correctly. In most churn prediction studies, session-related features, such as the number of logins, playtime, and the average time between games, were utilized as critical features that could indicate the extent to which players were engaged in a game [for example, 18,10]. Further studies added features of different kinds, such as social activities [25,31], achievements [8], and payments [51,52]. Most of the studies rely on machine learning-based approaches to deal with large and complex datasets. Consequently, the majority establish correlation rather than causation due to a lack of consideration of endogeneity problems. Recently, Xiong et al. [49] applied explainable artificial intelligence (XAI) to give interpretability to the prediction model and find causes of churn; still, the method assures less debiased results than causal analysis.

Other studies have focused on the experience or status that influences players' churn using statistical modeling or causal inference approaches. Lomas et al. [35] showed more trials and playtime when the players faced easier challenges. Sarkar et al. [39] also confirmed that difficulty influenced player engagement; matchmaking-based and difficulty-based ordering of tasks led to significantly more attempted and completed levels than random ordering. Achievements, such as kill points and quest completion, and social relationships, such as friend and guild interactions, affected the players to remain active and progress to higher levels [37,24,4]. Banerjee et al. [5] showed that currencies spent and equipment upgrades have positive effects on player activeness and engagement.

2.2. Multiplayer games and competitions

The development of online games has increased between-player interactions over the past decade. Consequently, competition and in-game social activities have become essential motivators for playing games [23]. In competition, players (or teams) attempt to block each other's attempts to ultimately win and obtain a higher reward [47]. An early stage study [46] suggested that players are more likely to enjoy playing a video game when confronted with competitive elements (e.g., monsters). Although there are concerns about competitive games (e.g., violence and aggression), numerous studies have shown that moderate competition significantly contributes to player motivation in various game genres, such as multiplayer online battle arenas, online board games, and online collectible card games [22,50,44].

2.3. Matchmaking

Studies on player experience have agreed that difficulty-adjusted matches are better than random matches. Matches in which players are equally skilled increase the frequency and duration of play [34] and are favored by players [43]. Therefore, typical matchmaking methods focus on *fair* matches, so the players can have the same chance of victory. A classic system, called the Elo rating, has been proposed as a probabilistic model for skill-based matchmaking [14]. Based on this model, researchers and game developers have improved matchmaking systems. For example, the Glicko system [15] and Trueskill system [19], enhance the accuracy and adaptability of skill ratings to provide players with the best experience in terms of fairness.

However, it is difficult to determine whether players prefer a fair match to an easy one. Some studies show that players spend more time and attempt more matches when they compete with equally skilled players [34,41]. However, Liu et al. [34] found that players felt more enjoyment when competing with lower-skilled opponents, and Lomas et al. [35] demonstrated that players were more engaged when they designed an easier educational game. We anticipated that players are less likely to churn when the level of match difficulty decreases.

H1. Being matched with a weaker opponent negatively influences churn and vice versa.

Additionally, we expected that unsuitable matchmaking would make players churn, as Tsai [43] showed that random matching lowers players' perceived enjoyment. In other words, players will churn more when their opponents' skill levels vary than when their opponents have a consistent skill level.

H2. Greater variance in skill between matched opponents positively influences churn.

2.4. Player performance

In addition to matchmaking, match results are also motivators because the main reason for competing is winning and being the most skilled person playing the game [23]. This motivation has been observed in various game genres with competitive content [44]. The desire to win is related to competence [21]. The authors found that players feel competent if they succeed in the given challenges of video games, which induces greater flow. When competing with others, the desire to continue playing can be reinforced [7]. Their study involved a case study regarding a popular massive multiplayer online role-playing game (MMORPG), World of Warcraft, where users playing in player-versus-player servers (mainly focused on players' confrontations) were more motivated by competition than users playing in player-versus-environment servers (mainly focused on challenges against non-player characters). Therefore, player's recent match results, such as win, lose, and draw, have become a significant factor in churn prediction in a competitive game [11]. To prove and measure this causal effect, we hypothesized the following:

H3. A higher win rate negatively influences churn.

Another interesting form of player performance is a streak, which this study defines as consecutive wins or losses. Many studies in sports have asserted that streakiness is a phenomenon in which continuous success leads to a greater chance of subsequent success and vice versa [6]. For example, Csapo et al. [12] discovered that athletes of the National Basketball Association were more confident and attempted more difficult shots when they had scored consecutive shots, resulting in better performance.

Kou et al. [28] conducted qualitative research on how online game players perceive streakiness. They argue that streaks are not only mere coincidences, but also a psychological phenomenon that can impact player experience. Many players in their study thought that their mentality and performance caused streaks to occur. Interestingly, in their study, even players who experienced winning streaks expressed negative emotions. The users lost confidence in the matchmaking system because they perceived that it intentionally created streaks. The authors suggested a different view from studies in sports that showed winning streaks led to better player experiences [1,6]. As different theories exist, we proposed separate hypotheses for winning streaks and losing streaks and tested them:

H4a. Experiencing more winning streaks negatively influences churn.

H4b. Experiencing more losing streaks positively influences churn.

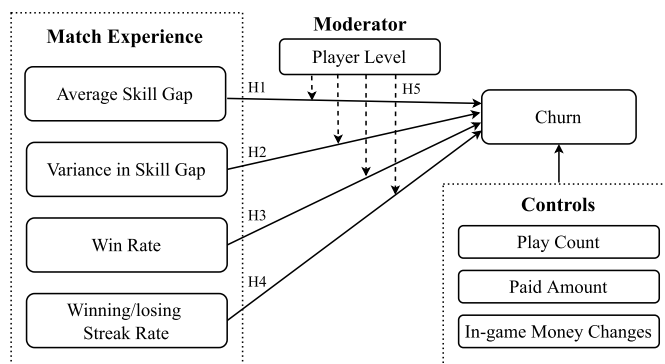


Fig. 2. Research model to estimate how match experience affects player churn.

2.5. Moderating effects of in-game phase change

Among the studies on player motivations, Park et al. [37] found that user retention varies depending on players' in-game phases in the case of MMORPG. They argued that achievement features (e.g., gained items, achieving the next level) highly affect player retention in early-to-mid in-game phases. In contrast, social relationships (e.g., number of friends) become more critical for player retention during the final in-game phase (i.e., the maximum level). Earlier research also observed that players' social activities increased at later levels because their available content became group-oriented in MMORPG [13]. This implies that it is necessary to consider players' in-game stages when investigating the causes of churn. Therefore, we proposed the following hypothesis to examine how in-game phases change the effects of the match experience:

H5. Players' in-game phases moderate the relationships mentioned in H 1–H 4b.

Fig. 2 depicts our hypothesized model. The model also includes three types of control variables: play count, paid amount, and in-game money changes.

3. Method

3.1. Game introduction

This research examined the match data of Everybody's Marble, a mobile multiplayer online casual board game developed by Netmarble. It was released in June 2013 and has been extremely popular in Korea, with more than 40% of the population having downloaded it due to its straightforward gameplay and familiar rules. The game is similar to the classic board game, Monopoly, as seen in Fig. 3. Two to four players are matched either 1:1 or 2:2 and roll dice to move around the game board. Players can buy cities and collect tolls when other players pass through them. Constructing buildings in a city increases the tolls. The game is won when all opponents are bankrupt due to paying tolls.

This game differs from Monopoly in that players can significantly improve their chances of winning by obtaining better items. Players can purchase these items with in-game currency earned by playing matches and completing secondary activities. Items provide various benefits to the owner during a match, such as reducing the cost of tolls, taking money from opponents, and even increasing the likelihood of rolling the desired number on the dice.

The game's matchmaking system pairs players based on their internally measured skills, known as matching points (MP). When a player requests to start a match, the system searches for another player who is waiting for a match and has MP similar to the requester. When the match is over, the system updates the MP of the players involved based on the outcome of the match. The winner gains and the loser loses MP. The number of changing points are determined by the company's personalized Elo rating system. The MP of the players in this game is positively correlated with their total playing time. Those who have played more matches have higher levels, earn more in-game currency, and purchase better items, which gives them a strong advantage.

It would be reasonable to assume that matchmaking quality and players' performance are essential components of this game. The game incentivizes players to strive for better performance by assigning ratings and providing corresponding rewards. According to posts from the game's official community, players desire to face easier opponents and are displeased with unfavorable matches. Furthermore, the game's secondary content also assists players to improve their performance by rewarding them with currencies for item purchases.

3.2. Data collection

The game publisher provided us with anonymized match records of 1,102,279 users from November 24, 2018, to January 20, 2019. It contains comprehensive information on player activities and matches, such as skill points for matchmaking, matching participants, and matching results.



Fig. 3. A game screen of Everybody's Marble: the figure shows a screen of the global version for better understanding [33].

Table 1

The panel data summary. Observations were aggregated daily, so there are (# player $\times$ # active date) observations.

Players	Observations	Observations per player			Date	Churn	Play count	Average skill gap	...
		Mean	Min	Max					
262,698	1,587,737	6.0	2	42	2018-11-24	0	14	8.67	
					2018-11-25	0	3	3.31	
					2018-11-28	1	4	9.89	

(a) The number players and observations in the panel data

(b) An example of the panel data of player A, who played for three days

We created panel data that contained player status and daily match experience. We set the start of the day not at midnight, but at 5 a.m. because this was when the users played the least. As many people play games around midnight, we wanted to keep continuous activities as uninterrupted as possible while the variables were aggregated. As this study aimed to determine the reasons for churn, we targeted players who left the game in our log period. We defined churn as 14 or more days of absence (see Section 3.3 for detailed information). Therefore, the last 14 days of our game log were the cutoff point that determined whether players left the game. Next, we obtained the data for 42 days (from November 24, 2018, to January 6, 2019) for panel analysis. As we designed a player-within fixed-effects model, observations that met the following conditions were used to fit our model. (See Section 3.4 for why we used this model.)

- 1) A player should have played for at least two days in the analysis period. If not, it was not possible to measure within-player variation over time.
- 2) A player's churn status should have changed during the analysis period. This means a player should have left the game for 14 or more consecutive days during the period.

After filtering according to the above conditions, our panel data covered 1.5 million observations logged from 262,698 players. The conditions excluded players who logged into the game but did not play any matches and those who played only one day, as there were not enough observations to analyze the match experience. The players who did not churn during the period were also beyond the scope of our study. Table 1 summarizes the panel data. The players played the game for about six out of 42 days on average,¹ which means the panel data is unbalanced. See Fig. A.1 for the distribution of the observations.

3.3. Measures

We collected players' match, access, and money logs and designed variables that could describe their match experience and current status. Table 2 shows the variables and their descriptive statistics. See Fig. A.2 for the distributions across levels and days.

The average and variance in skill gaps are related to matchmaking quality. Mostly, the matchmaking system performed balanced matchmaking because the MP differences between players and opponents were close to a normal distribution (mean = -0.1, SD = 17.9, when MP is normalized to the 0–100 range) during the analysis period. The average win rate was 50.7% when players met opponents with almost the same MP. However, some players encountered opponents with very different skill levels throughout the day. The standard deviation of skill gaps per day was more than 15 points in 14.7% of the cases. We had the following two independent variables to test H 1–H 2 (defined in Section 2.3):

- 1) *Average skill gap*: The average skill gap a player experienced during the day. We measured the skill gap as the opponent's MP (the average MP if there are many opponents in one match) minus the player's MP. Therefore, a player meets a stronger opponent if the skill gap is a positive number; a player meets a weaker opponent if the skill gap is a negative number.

¹ Note that newly registered users during the period were included.

Table 2
Descriptive statistics of variables.

Variable	Mean	SD	Min	Max
Churn status (per day)				
Churn	0.17	0.38	0	1
Match experience (per day)				
Average skill gap	6.24	16.45	−90.45	99.98
Variance in skill gap	7.25	7.74	0	74.23
Win rate	50.94	35.45	0	100
Winning streak rate	9.13	18.03	0	98.82
Losing streak rate	7.63	15.65	0	98.65
Player status (per day)				
Level	57.97	31.99	1	125
Play count	6.40	8.20	1	365
Paid amount	0.85	7.39	0	100
Cash earned	5.30	2.12	0	9.29
Cash spent	3.68	3.08	0	10.89
Cash remained	6.14	2.19	0	12.84
Gold earned	11.96	2.63	0	18.22
Gold spent	10.86	3.50	0	18.14
Gold remained	13.51	1.69	0	20.14

Note. SD: standard deviation

2) *Variance in skill gap*: The standard deviation of skill gaps a player experienced during the day.

The other dependent variables are related to player performance. They were designed to test H 3–H 4b (defined in Section 2.4). Win rates show players' cumulative performance during the day. Additionally, streakiness can impact user experience, according to Kou et al. [28]. In their study, the players surveyed defined a streak in several ways: consecutive wins or losses (62% of the players), sudden changes in rank (36%), experiencing wins or losses only (15%), and an imbalanced win rate (2%). We used the most frequent definition, consecutive wins or losses. We defined a streak as beginning from the 3rd match in a losing (or winning) match sequence. When players lose three or more matches consecutively, the game system suggests a practice match with an artificial intelligence (AI) player to encourage them by providing a winning experience. However, there is no special action in the case of consecutive wins, but the threshold is set to be the same to balance with the case of consecutive losses.

- 1) *Win rate*: The cumulative win rate by day (%).
- 2) *Winning streak rate*: The rate of matches in winning streaks of all matches by day (%). For example, let us assume that player A had 10 matches in a day d , and the player's win-lose sequence is 0-1-1-1-0-1-0-0-0-0 where 1 is win, and 0 is lose. Then the winning streak rate is 10% because there is one winning streak 1-1-1, and the counting starts from the third match.
- 3) *Losing streak rate*: The rate of matches in losing streaks of all matches by day (%). In the same example of player A, the losing streak rate is 20% because there is one losing streak 0-0-0-0, and the third and fourth matches are counted.

As mentioned in Section 2.1, we anticipated that players' churn reasons differ depending on their in-game phases. The in-game level is appropriate for inferring how long users have played; therefore, we added it as a moderator.

- 1) *Level*: The final level of the day.

We had several control variables to minimize selection bias. We attempted to measure players' dedication to the game by their playing frequencies, real money paid, and in-game monetary transactions. These features are highly correlated with churn, and this correlation has been verified many times in several churn prediction studies [18,38,32,52].

- 1) *Play count*: The total number of matches played per day.
- 2) *Paid amount*: The amount of real money paid during the day. Given the game company's compliance, this variable was measured as percentile ranks per day.

We log-transformed the following in-game monetary variables because their histograms were extremely skewed to the right:

- 3) *Cash earned/spent/remained*: The amount of in-game cash earned/spent/remained during the day. Cash plays an important role in enhancing character performance, and opportunities to obtain it are limited (e.g., microtransactions, limited contents).
- 4) *Gold earned/spent/remained*: The amount of in-game gold earned/spent/remained during the day. Gold plays a supporting role in enhancing character performance and can easily be obtained through match rewards.

Finally, we were interested in how the above variables affected churn. Thus, churn was the last variable. Generally, churn is measured by user absence periods. Previous researchers have had different criteria to determine the absence period. For example, 7,

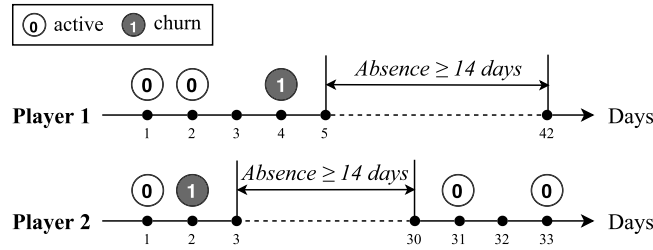


Fig. 4. Some examples of churn labeling. A circle (○) means a day the player played the game. Player 1 had not played the game since Day 5; therefore, Day 4—the last day the user played before churn—was labeled as *churn*(1). Player 2 (returner) left the game on Day 3, but the user started playing again on Day 31. Therefore, Day 2 was labeled as *churn*(1), and the dates from Day 31 were labeled as *active*(0).

14, 28 days, or even 13 weeks [38]. The period usually depends on the game genre and operation policy of the publisher. According to the publisher's recommended policy, our study defined churn as a user not playing the game for 14 days or more. With this criterion, at least 80% of the churners in our data did not return to the game for more than a month. Fig. 4 shows two examples of churn labeling.

1) *Churn*: If a user has not played any matches for the next 14 days, the value is 1. Otherwise, the value is 0.

3.4. Analytic strategy

3.4.1. Possible bias in game data

The ideal experiment to examine causal relationships is to assign treatments randomly. However, it is often impractical to conduct it on live online games. It is preferable to investigate pre-collected game logs because they are easy to obtain, advantageously large, and standardized. They contain a large multitude of players and in-game actions. They can be easily transformed into panel data (also called longitudinal data) that tracks players' experiences over time.

The problem is that it is challenging to make a causal inference from game logs. Usually, game logs are non-experimental data. User heterogeneity and time heterogeneity can easily arise because game logs have limitations in collecting data unrelated to gameplay. For example, the publisher of our studied game did not collect users' personal information to protect their privacy; therefore, factors such as gender, age, and job, could not be controlled in our research model. Moreover, logging dates can cause time-variant heterogeneity. We expected unobserved time heterogeneity in two aspects. First, there are more concurrent users on weekends, holidays, and event periods. The number of concurrent users influences matchmaking results, as the matchmaking system searches for an opponent with a similar MP within a time constraint. Therefore, players are more likely to meet an opponent with a small skill gap if there are more concurrent users. This means that our matchmaking-related variables (the average and variance of skill gaps) can be biased over time. Second, players' activity patterns can vary during events or after version updates due to content changes, which can also cause biased estimates.

3.4.2. Two-way fixed-effects model

To account for user and time heterogeneity, we used a two-way fixed-effects estimation [9]. The dependent variable y_{it} (churn) is binary; therefore, we considered the following fixed-effects logistic regression model (Eq. (1))

$$\text{logit}(\Pr(y_{it} = 1)) = \beta X_{it} + \alpha_i + \gamma_t + \epsilon_{it} \quad (1)$$

for user i and date t . X_{it} denotes the vector of explanatory variables and β is the vector of the corresponding slopes of log-odds. α_i denotes the user fixed-effect, γ_t is the time fixed-effect, and ϵ_{it} is the error term. The fixed-effects model considers α_i and γ_t not as random variables, but as parameters to be estimated. This method can control heterogeneity issues by measuring the effects while fixing both user and time.

We were also interested in whether a player's in-game phase is a moderator, as assumed in H 5. We divided the player levels into five phases, as shown in Table 3. The in-game phase was included as a dummy variable (Ph_{it}) interacting with other match-related variables. Thus, our other estimation equation (Eq. (2)) was

$$\text{logit}(\Pr(y_{it} = 1)) = \beta_1 X_{it} + \beta_2 X_{it}^m Ph_{it} + \beta_3 Ph_{it} + \alpha_i + \gamma_t + \epsilon_{it} \quad (2)$$

where X_{it}^m denotes the five variables under "Match experience" in Table 2. Using this equation, we determined whether the in-game phase changes the effects of the match experience on churn.

We should be careful when interpreting the coefficients of the logistic regression model. The estimated coefficient shows how much the log odds ratio changes when the explanatory variable is increased by one unit. The odds ratio is the ratio of the churn probability to the retention probability. For instance, if the coefficient is 1, the odds ratio increases to approximately $e^1 \approx 2.7$ when the variable is increased by 1. This is not easy to understand as it does not show the absolute change in the churn probability.

Table 3

The in-game phases and the number of corresponding players and observations.

Phase	Level range	Players	Observations	Avg. play count
Ph0	1–25	82,994	274,609	112
Ph1	26–50	93,862	466,363	508
Ph2	51–75	57,089	366,886	868
Ph3	76–100	38,496	283,044	2,074
Ph4	101–125	19,650	196,835	6,759

Note: 1. Avg. play count: The average number of matches required to enter the next phase (or to achieve the highest level in case of the last phase Ph4). 2. The sum of players slightly exceeded the total number of players because some players went through multiple in-game phases during the analysis period.

To explain the results more intuitively, we estimated the average semi-elasticity (ASE) of the coefficients. *Semi-elasticity* measures a percentage change in an outcome for a unit change in an explanatory variable. Existing studies show that fixed-effects logit models are capable of consistently estimating this value [27,26]. Specifically, semi-elasticity e is defined as Eq. (3)

$$e_{it} = \frac{\partial \ln \Pr(y_{it} = 1 | x_{it}, \alpha_i)}{\partial x_{it}} = \beta \frac{1}{1 + \exp(\beta x_{it} + \alpha_i)} \quad (3)$$

and the ASE can be computed as Eq. (4)

$$ASE = E[e_{it}] = \beta(1 - E[y_{it}]) \quad (4)$$

which can be estimated using $\hat{\beta}$ and $\bar{y} = \frac{\sum_{i=1}^n \sum_{t=1}^T y_{it}}{nT}$. For example, if the ASE is 1, the churn probability increases by 1%, on average, when the variable increases by 1. Note that the unit of the outcome change is not a percentage point but a percentage.

4. Empirical analysis and results

4.1. Covariate analysis

Before running the model, we conducted a covariate analysis to check that our observation dataset was not too imbalanced. We examined the distributions of the variables and their bivariate relationships to check the difference in the covariates between the players who churned early and those who churned late. The covariates described by histograms and scatter plots showed that their distribution overlapped for both groups, implying that the dataset was well-balanced (see Fig. A.3). The Pearson correlation coefficients differed by less than 0.08 between the two groups. Exceptionally, the level variable showed a slightly different distribution; the players who churned early tended to have lower levels than those who churned late. Our study solved this issue by Eq. (2) model, calculating the effect of match experience variables under divided level intervals.

4.2. Model specification

We conducted the Hausman test to check whether adopting the fixed-effects model was appropriate [3]. The alternative hypothesis of the test was that individual characteristics were correlated with the regressors, indicating that the user fixed-effects model was preferred. The test result was $\chi^2_{(13)} = 215,753, p < .001$, supporting the alternative hypothesis. We also conducted the Wald test. The test result $\chi^2_{(41)} = 87,068, p < .001$ showed that the coefficients for the dates were significant, meaning that the time fixed-effects were required. The likelihood-ratio (LR) statistics of the model was $\chi^2_{(13)} = 68,796, p < .001$, implying that the model was a good fit.

To check and compare the models' goodness of fit, we measured McFadden's R^2 [36] and the Bayesian information criterion (BIC) [40], as shown in Table 4 and Table 5.

4.3. Main results

Table 4 shows the two-way fixed-effects logit estimation results of Eq. (1). The positive coefficients suggest that churn probabilities increase when the independent variables increase (and vice versa with the negative coefficients). The estimated ASE values are also stated, together with the coefficients for easier interpretation.

The results showed that our matchmaking quality-related variables significantly influenced churn. The average skill gap experienced per day had an ASE of 0.0013, which means that when it increased by 1, the churn probability increased by 0.13 %. The variance in skill gaps experienced per day had an ASE of 0.0023, which means that when the variance increased by 1, the churn probability increased by 0.23%. Both variables showed significant and positive effects on churn, supporting H 1 and H 2.

Regarding the player performance variables, the win rate and winning streak rate significantly influenced churn, whereas the losing streak rate did not. The win rate by day had an ASE of -0.0031, decreasing the churn probability by 0.31 % when the rate

Table 4
The effect of match-based variables on churn (Eq. (1)).

Variable	Coefficient	ASE
Match experience		
Average skill gap	0.0016*** (0.0002)	0.0013
Variance in skill gap	0.0027*** (0.0004)	0.0023
Win rate	−0.0037*** (0.0001)	−0.0031
Winning streak rate	−0.0104*** (0.0002)	−0.0086
Losing streak rate	0.0001 (0.0002)	0.0001
Controls		
Play count	−0.0713*** (0.0009)	−0.0591
Paid amount	0.0001 (0.0005)	0.0001
Cash earned	−0.0890*** (0.0015)	−0.0737
Cash spent	−0.0236*** (0.0012)	−0.0195
Cash remained	0.0181*** (0.0023)	0.0150
Gold earned	−0.0158*** (0.0011)	−0.0131
Gold spend	−0.0145*** (0.0009)	−0.0120
Gold remained	0.0897*** (0.0032)	0.0743
Pseudo- R^2	.272	
BIC	587,519	

Note. *** $p < .001$, ** $p < .01$, * $p < .05$; the standard errors are in parentheses.

increased by one percentage point. The winning streak rate by day had an ASE of -0.0086, decreasing the churn probability by 0.86 % when the rate increased by one percentage point. Thus, H 3 and H 4a were supported. However, the losing streak rate did not significantly affect churn ($p = .773$), thus rejecting H 4b.

The estimates of the control variables also showed some interesting findings. Play count showed a strong negative relationship with churn. Paid amounts are known to be strong predictors of churn, but they did not show a significant relationship with churn in this study ($p = .853$). Regarding in-game monetary variables, cash and gold showed similar characteristics. Both money earned and spent had significant negative relationships with churn, while money that remained had a significant but positive relationship.

4.4. Moderating effects of in-game phase changes

The estimation results for the Eq. (2) are described in Table 5. As previously mentioned, the second model includes the interaction terms of the match experience variables and in-game phases. Additionally, Table 6 shows the standardized coefficients, as the match variables were measured on different scales.

The in-game phases had no moderating effects on the average skill gap and variance in skill gaps. The table shows that all the corresponding interaction terms relevant to the variables failed the significance test ($p > .05$). The in-game phases marginally moderated winning streak rates. When players belonged to Phase 1, the effect size became slightly larger, but all other phases had no significant effects.

However, the in-game phase showed a notable moderating effect on the win rate and losing streak rate. The interactions relevant to the two variables exhibited significant effects in every phase. We plotted Fig. 5 to illustrate the effect changes in the in-game phases. Each point in the figure denotes the ASE value when the corresponding phase dummy is one (base ASE + treatment * level phase ASE). The size of the win rate's ASE tended to increase through the in-game phases. The coefficient sign of the losing streak rate changed through the in-game phases. It showed negative impacts until Phase 1 but showed positive impacts from Phase 2.

Also, Table 5 includes the main effects β_3 of the in-game phase variable. All phases had significant effects ($p < .001$), indicating that the players had different intercepts depending on their levels. Note that the coefficients increased through the in-game phases because the in-game phase is a monotonically increasing value over time in any given player.

4.5. Robustness checks

4.5.1. Comparing periods before and after a game update

To ensure the robustness of our model, we first considered game updates that change in-game content. A game update can affect players' engagement because it influences pre-established rules of competition by adding new items or ways to compete with others [20]. There was a regular game update that added a new type of board game and a new event during the analysis period on December 12, 2018. We divided the data into two time windows, the period before and after the update, and checked if the effect of the match experience changed. The results are shown in Table B.1 in the Appendix. Most of the signs and magnitudes of the coefficients of the match experience variables were consistent with our main results (in Table 4). The average and variance of skill gaps became insignificant in the case of the period before the game update; however, the coefficients were close to the main results. This indicated that our key factors explained churn well, even after the update, without any major changes.

Table 5
Moderating effects of in-game phase change (Eq. (2)).

Variable	Coefficient	ASE
Match experience		
Average skill gap	0.0024*** (0.0004)	0.0020
Average skill gap * Ph1	−0.0009 (0.0006)	−0.0008
Average skill gap * Ph2	−0.0011 (0.0007)	−0.0009
Average skill gap * Ph3	−0.0004 (0.0008)	−0.0003
Average skill gap * Ph4	−0.0014 (0.0010)	−0.0011
Variance in skill gap	0.0038** (0.0011)	0.0031
Variance in skill gap * Ph1	−0.0024 (0.0013)	−0.0020
Variance in skill gap * Ph2	−0.0012 (0.0014)	−0.0010
Variance in skill gap * Ph3	−0.0028 (0.0015)	−0.0024
Variance in skill gap * Ph4	−0.0010 (0.0017)	−0.0009
Win rate	−0.0023*** (0.0002)	−0.0019
Win rate * Ph1	−0.0016*** (0.0002)	−0.0013
Win rate * Ph2	−0.0022*** (0.0003)	−0.0018
Win rate * Ph3	−0.0020*** (0.0003)	−0.0016
Win rate * Ph4	−0.0022*** (0.0004)	−0.0018
Winning streak rate	−0.0096*** (0.0004)	−0.0079
Winning streak rate * Ph1	−0.0012* (0.0005)	−0.0010
Winning streak rate * Ph2	−0.0009 (0.0006)	−0.0007
Winning streak rate * Ph3	−0.0004 (0.0006)	−0.0003
Winning streak rate * Ph4	−0.0002 (0.0008)	−0.0001
Losing streak rate	−0.0019*** (0.0005)	−0.0016
Losing streak rate * Ph1	0.0016** (0.0006)	0.0013
Losing streak rate * Ph2	0.0024*** (0.0007)	0.0020
Losing streak rate * Ph3	0.0036*** (0.0007)	0.0030
Losing streak rate * Ph4	0.0044*** (0.0009)	0.0037
In-game phase		
Ph1	2.3532*** (0.0453)	1.9483
Ph2	3.8382*** (0.0665)	3.1777
Ph3	5.1312*** (0.0912)	4.2482
Ph4	6.3449*** (0.1260)	5.2531
Controls		
Play count	−0.0742*** (0.0009)	−0.0615
Paid amount	0.0000 (0.0005)	0.0000
Cash earned	−0.0871*** (0.0015)	−0.0721
Cash spent	−0.0226*** (0.0012)	−0.0187
Cash remained	0.0185*** (0.0024)	0.0153
Gold earned	−0.0166*** (0.0011)	−0.0138
Gold spend	−0.0145*** (0.0009)	−0.0120
Gold remained	0.0874*** (0.0032)	0.0724
Pseudo- R^2	.280	
BIC	581,034	

Note. *** $p < .001$, ** $p < .01$, * $p < .05$; the standard errors are in parentheses.

Table 6
The standardized coefficients of match experience variables (Eq. (2)).

Variable (Var)	Standardized coefficient				
	Var (base)	Var * Ph1	Var * Ph2	Var * Ph3	Var * Ph4
Average skill gap	0.0395***	−0.0148	−0.0181	−0.0658	−0.0230
Variance in skill gap	0.0294**	−0.0186	−0.0093	−0.0217	−0.0077
Win rate	−0.0815***	−0.0567***	−0.0780***	−0.0709***	−0.0780***
Winning streak rate	−0.1731***	−0.0216*	−0.0162	−0.0072	−0.0036
Losing streak rate	−0.0297***	0.0250**	0.0376***	0.0563***	0.0689***

Note. *** $p < .001$, ** $p < .01$, * $p < .05$

4.5.2. Subsample analyses in terms of playing activeness and newcomers

Furthermore, we performed subsample analyses to check if match experience affects churn differently in specific player groups. Players might experience matches differently depending on their engagements; therefore, we split them into more active and less active groups. Technically, we measured players' activeness with the daily average play count and divided them into the top 50 percentile ("high activeness") and bottom 50 percentile ("low activeness") groups. Also, players that are new to the game might perceive variables differently; therefore, we split them into another set of groups that contained those who had newly joined during the log period ("new players") and those who had joined before the log period ("existing players"). The estimated results are described in Table B.2. Overall, the directions of match experience effects were consistent with our main results. There were some small

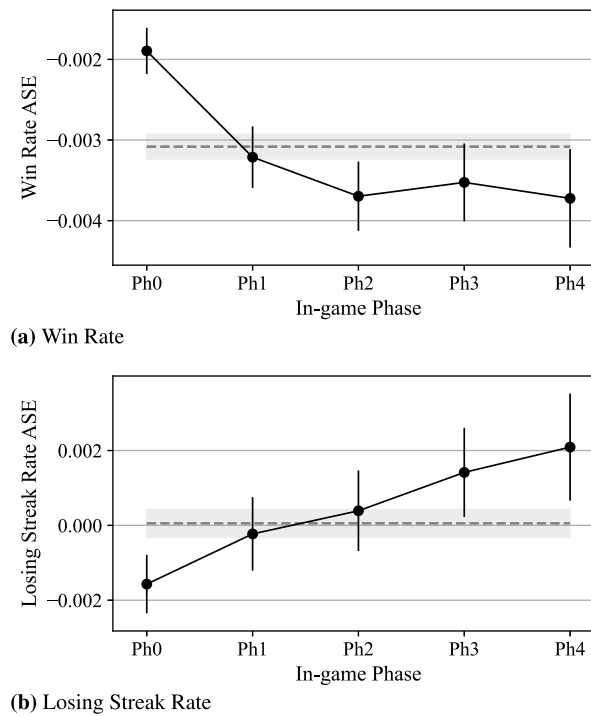


Fig. 5. The moderating effects of in-game phases on churn. A win rate and a losing streak rate have different effects on churn depending on the phases. The black line shows the ASE and its 95% confidence interval with the interaction, while the dotted line and the grey area show the ASE and its 95% confidence interval without the interaction.

changes; the variance in skill gaps of the high activeness group and the average skill gap of the new players became insignificant. The losing streak rate became significant for the low activeness group.

5. Discussion

5.1. Matchmaking experience

First, our results related to the average skill gap indicate that players churn less when they compete with easy opponents and churn more when they compete with difficult opponents. This finding supports the claim that players are more engaged when a game is easier [35]. However, it contradicts another argument that players enjoy a fair match [34]. A possible explanation for this could be the value range of the observed skill gaps. As our game system already provides a balanced match to some degree, there are few extremely unbalanced matches in the data. Therefore, our model may not be sufficiently tested for extremely easy (or extremely difficult) matches. Nevertheless, our results support that players prefer easier matches in a moderate range of a skill gap. Second, another finding is that the variance in skill gaps positively influences churn probability, therefore reducing the variance may help retain players. This finding is consistent with the earlier study of Li et al. [32] which suggests that an adjusted variance in perceived difficulty in a solo puzzle game reduced churn.

5.2. Match results and streaks

Our results revealed that both the win rate and winning streak rate negatively affected churn, which improves user retention. As expected, our findings supported the idea that winning is one of the key motivations in competitive games [23]. However, explaining the results of the winning or losing streak requires further attention. As mentioned in the literature review, two aspects explain streaks: 1) the theory that a winning streak fosters more confidence and raises the chance of the next win in sports [12], and 2) the theory that both winning and losing streaks negatively impact players in online games [28]. Our results on streaks indicate that a winning streak positively affects the player experience. Nonetheless, there are possible explanations for this discrepancy between sport and online gaming research. Kou et al. [28] used a qualitative method to examine players' subjective perceptions. The streak that the participants mentioned in their study averaged over ten consecutive wins or losses, which is considerably longer than the average of five wins or losses observed in our data. Combining these findings, we concluded that a short winning streak appears to provide positive experiences, but might become a negative experience when players undergo an inordinately long streak.

5.3. Characteristics of different in-game phases

The model that accounts for the in-game phases (Eq. (2)) better explains churn than the model that does not (Eq. (1)). By adding the main effects and interaction terms of the in-game phases, the R^2 value increased from .272 to .280, and the BIC value decreased from 587,518 to 581,034. Thus, it is reasonable to assume that the effects vary across the players' phases, which adds further evidence to consider players' lifecycles, together with churn analysis studies [37] and churn prediction studies [30,38,32]. In particular, the performance variables (win rate and losing streak rate) had different effects on each in-game phase, while other variables (average skill gap, variance in skill gaps, and winning streak rate) mostly had consistent effects. The performance variables are more visible to the players than the matchmaking variables, as the matching points are not disclosed.

The win rate affected the advanced players more than the beginners. The beginners (Ph0) appeared to be less concerned about achieving low win rates. One possible reason is that players who are not experienced feel less competent, regardless of the level of challenge [21]. The win rate became important when the players were accustomed to the game. An earlier study showed partially similar conclusions [37]. Analyzing a MMORPG, the latter study demonstrated that achievement affected churn more as players raised their levels until the medium-level phase. However, contrary to our results, social activities became much more important than achievement for the highest-level players. The most convincing explanation is the difference in game content across game genres. Typically, content before the maximum level in MMORPGs focuses on individual achievements, such as level-ups and item reinforcements. When players achieve the maximum level (in other words, when they have collected all achievement-related individual content), their activities often become group-oriented to form a party and tackle the most difficult quests [13]. However, the players in our game can compete with those with similar skills even after reaching the maximum level, which causes the effect of win rates to last.

Interestingly, the losing streaks impacted churn significantly when considering in-game phase interactions. Moreover, low-level players tended to remain when their losing streak rates increased. From a common sense standpoint, a losing streak should negatively affect a player, as demonstrated by [28]. However, their finding was only applicable to certain phases in our research. We inferred a reason: as the higher losing streak rates resulted in more long-losing streaks than multiple short-losing streaks, low-level players may have been more sensitive to the frequency of losing streaks than to their length. It is also worth noting that the effect size of the winning streak rate was notably larger than that of the losing streak rate in every in-game phase. This indicates that a winning streak can be more rewarding to players.

5.4. Comparison and interpretation of the coefficients

Although we interpreted the ASEs of the variables, it would be meaningful to compare their standardized coefficients as shown in Table 6. The standardized effect size of the win rate and winning streak rate (ranging from 0.08 to 0.19) are far larger than those of other variables (ranging from 0.00 to 0.04). This noticeable difference indicates that players value their win rates and win streaks over other factors, although we must be cautious not to interpret standardized coefficients as exact effects [16].

A slight change in the match experience factors might not influence churn considerably. However, it is worth avoiding excessively unbalanced matches. Based on Table 4, a 50-point increase in the average skill gap can increase the churn probability by 10%. If a player with a churn probability of 17% (the average churn rate of our dataset) undergoes such an experience, the player's churn probability will increase to 18.7%. It might seem like a small change, but for massive multiplayer games, even a 1.7% point increase in churn rate can be critical. Similarly, frequently biased matches should be avoided because a 30% point decrease in the win rate will result in an almost 1% point increase in the churn rate.

5.5. Implications and suggestions

Our study contributes to churn analysis in the online gaming universe by investigating data from a large-scale real game. Many previous studies in this field have aimed to accurately predict churn. Our study focused on the causes of churn. As churn is an indicator that reflects player engagement and can easily be measured, it can be useful in non-experimental research utilizing game logs if an appropriate approach is supported. We analyzed our results with experimental studies on player experience [for example, 35,34] or studies on other game genres [for example, 37], and explained the differences. Our work suggests that matchmaking, matching results, and streaks can significantly affect player experience, and the degree of the effect can change depending on players' in-game phases in competitive games.

This study also presents practical implications for competitive game developers in lowering their game churn rates. Our findings are closely related to matchmaking problems; they show that maintaining stable outcomes of matchmaking systems is necessary because the lower variance in skill gaps decrease churn. However, it can be problematic for developers to manage an individual's win rates or streaks. Intentional control over matches may have drawbacks because players can feel unreasonable [28]. Thus, it would be better to consider collateral strategies rather than direct intervention. For example, mid- to high-level players are relatively sensitive to their win rates. Therefore, game developers can adjust the matchmaking algorithm to match these players and opponents with smaller skill gaps, even though it may lead to a slightly longer wait time. Decreasing their losing streak rates, suggesting a match with an AI player, or providing guidance to enhance skills are preventive measures against churn. Moreover, low-level players are less concerned about winning or losing; therefore, focusing on out-match factors can be more helpful in lowering their churn rate.

5.6. Limitations and future directions

First, although this study used well-structured logs from the game server, the scope of the information that can be collected within a game log is rather limited. Whether players churn or not can be clearly determined through game logs, but it is difficult to measure the potential churn probabilities of active users that change over time. Future research could consider increasing the variety of variables, such as collecting players' demographics, out-game situations (e.g., day job, busyness), and perceived enjoyment after a match using a short pop-up survey or rating. The second limitation is that our study focused only on player-within effects. Our methodology could not cover players who had one observation or had not churned during the log period. In some cases, the effects on the population (e.g., an overall churn rate change by an event) would be meaningful rather than individual effects; therefore, future research could investigate this goal. Third, our study has certain limitations in verifying our findings outside our observation dataset. A carefully designed A/B test or a survey from actual game players would be appropriate for future studies to address this issue and enrich the findings.

Our study provides insights into how match-related experiences affect player churn in competitive games, but there is abundant space to investigate further causes of churn. Studies on the relationship between churn and other aspects such as social, game design, and cheating-related player experiences are required to identify additional reasons for game users' churn.

Ethics declarations

The Korea University Institutional Review Board waived the need for ethics approval and the need to obtain consent for this study, exemption number IRB-2023-0273 (KUIRB-2023-0306-01), because the authors used secondary data that was anonymized and included no personally identifiable information. This study complied with the Personal Information Protection Act requirements for research using anonymized data.

CRediT authorship contribution statement

Hyunjae Kang: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Conceptualization. **Changwoo Suh:** Writing – original draft, Validation, Methodology, Data curation. **Huy Kang Kim:** Writing – original draft, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data is not publicly available because the authors do not have permission to share the data.

Acknowledgements

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Appendix A. Panel data descriptions

As the raw game data used in this study is not publicly available, we outlined all the variables defined in Section 3.3. Fig. A.1 implies the distribution of the number of observations in each player's panel (left) and the distribution of churn status across days (right). Fig. A.2 describes the distribution of each variable across in-game phases and days. Most of the variables show similar distributions across days. However, the amount of cash earned and spent increased suddenly on December 12, 2018, because a new cash event had started.

Fig. A.3 describes the distribution of each variable and their bivariate relationships by histograms, scatter plots, and Pearson correlation coefficients. To compare the covariates between players who churned early and those who churned late, we divided the players into two groups based on the date they churned.

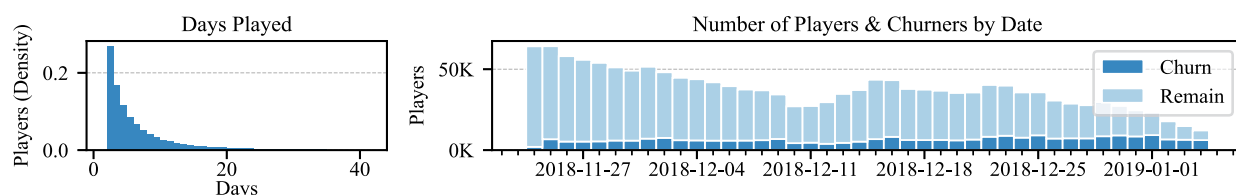


Fig. A.1. (Left) The distribution of players by the days played during the analysis period. (Right) The number of users who played the game and whether they churned on each date. The number of players decreased as the players at the beginning of the period churned as time passed.

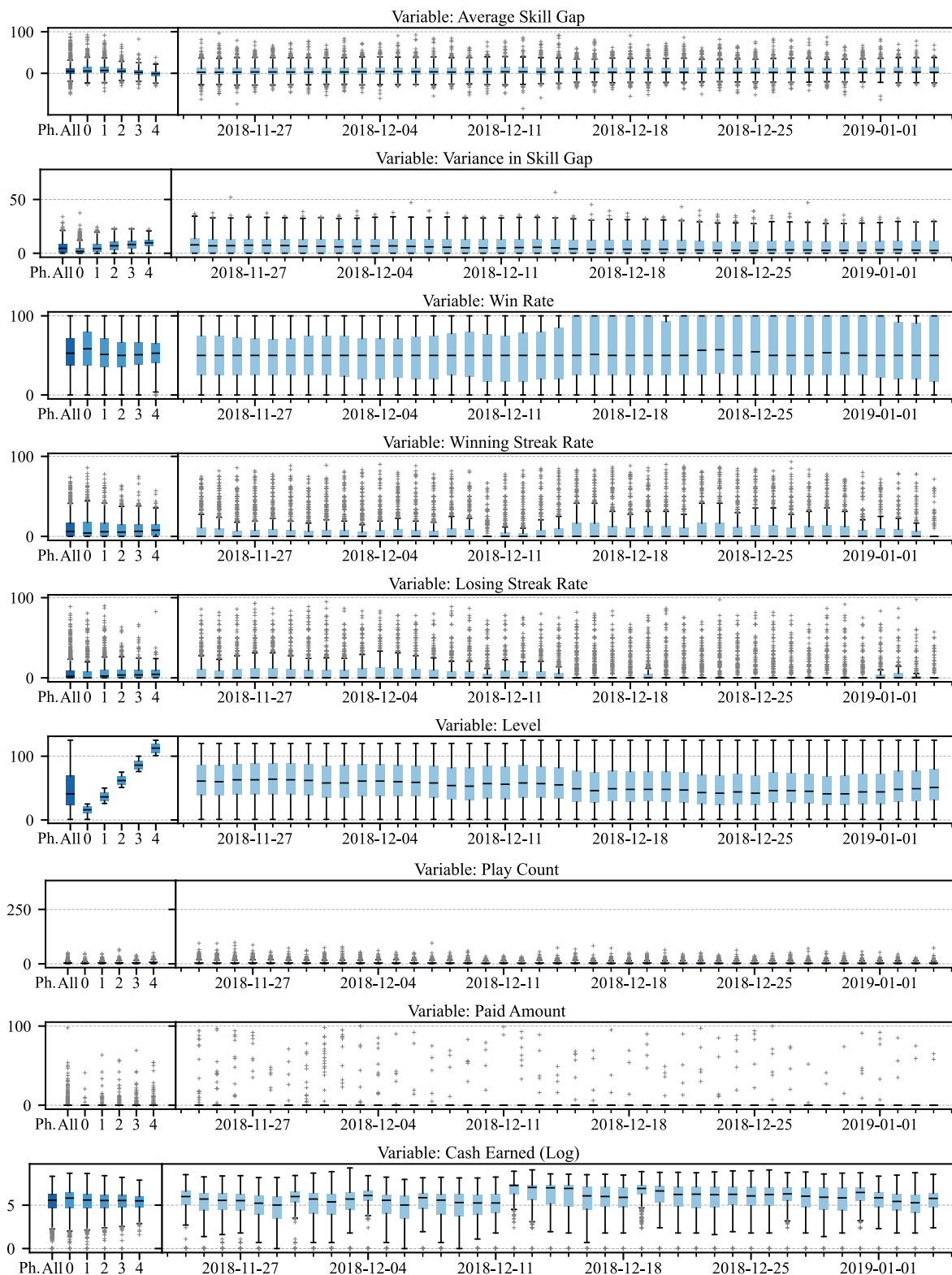


Fig. A.2. The distributions of all the variables by in-game phase (Ph. 0–4) and date. The play count and paid amount show a long-tail distribution; few players have extremely higher play counts or larger paid amounts than most.

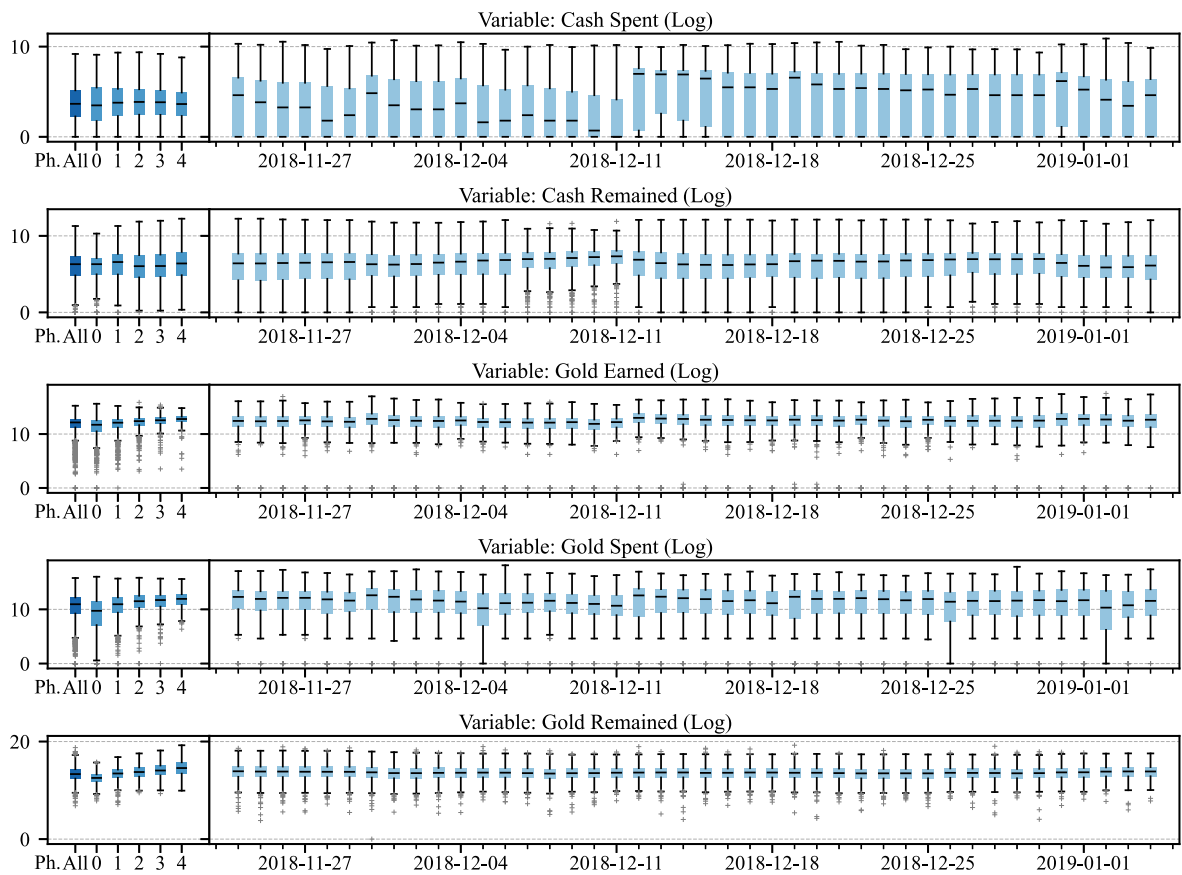


Fig. A.2. (continued)

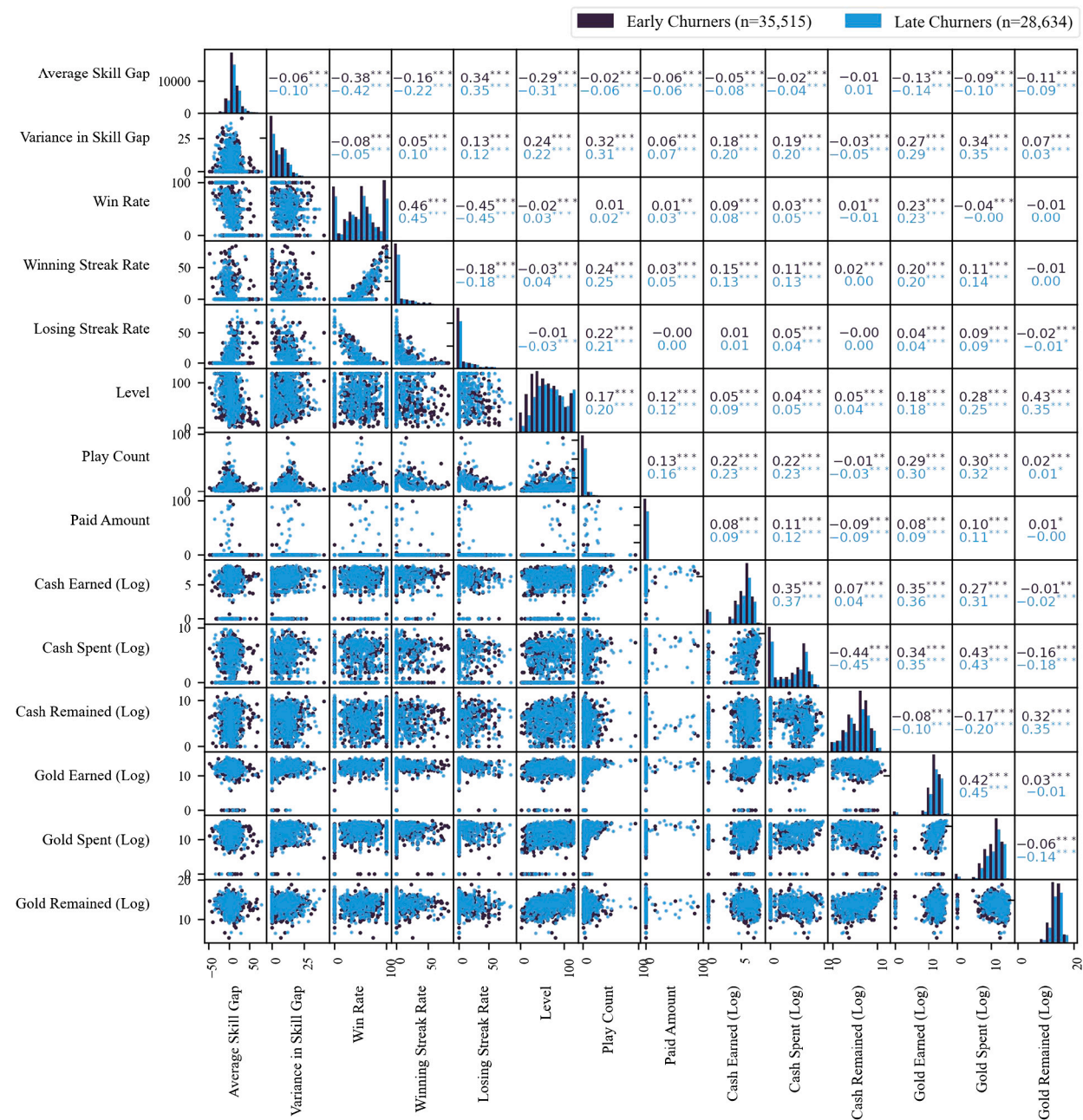


Fig. A.3. The scatter plot matrix of covariates of the players who played on the first date of the analysis period (November 24, 2018). The Early Churners group comprised those who churned in 1–20 days, and the Late Churners group comprised those who churned in 21–42 days. The main diagonal shows the histograms for each variable. The lower triangle shows the scatter plots for the bivariate relationship between variables. The upper triangle shows the Pearson correlation coefficients and associated p-values: *** $p < .001$, ** $p < .01$, * $p < .05$.

Appendix B. Robustness checks

Table B.1

Different time windows: splitting the data before and after the regular update on December 12, 2018.

Variable	Coefficient	
	Before the update	After the update
Match experience		
Average skill gap	0.0010 (0.0011)	0.0011* (0.0005)
Variance in skill gap	0.0039 (0.0020)	0.0039*** (0.0009)
Win rate	−0.0013** (0.0005)	−0.0020*** (0.0002)
Winning streak rate	−0.0063*** (0.0009)	−0.0069*** (0.0004)
Losing streak rate	−0.0001 (0.0010)	0.0004 (0.0005)
Controls		
Play count	−0.0197*** (0.0026)	−0.0394*** (0.0016)
Paid amount	0.0009 (0.0022)	0.0010 (0.0011)
Cash earned	−0.0394*** (0.0065)	−0.1118*** (0.0036)
Cash spent	−0.0039 (0.0059)	−0.0199*** (0.0025)
Cash remained	−0.0018 (0.0131)	0.0320*** (0.0051)
Gold earned	−0.0165** (0.0052)	−0.0090*** (0.0024)
Gold spent	−0.0140** (0.0047)	−0.0037 (0.0019)
Gold remained	0.0348* (0.0170)	0.0733*** (0.0066)
Players	94,557	144,271
Observations	401,846	643,914
Pseudo- R^2	.886	.668
BIC	27,867	125,370

Note. *** $p < .001$, ** $p < .01$, * $p < .05$; the standard errors are in parentheses.

Table B.2

Subsample analyses: dividing player groups in terms of playing activeness and newcomers.

Variable	Coefficient		Coefficient	
	High activeness	Low activeness	New players	Existing players
Match experience				
Average skill gap	0.0019*** (0.0004)	0.0019*** (0.0003)	−0.0004 (0.0009)	0.0018*** (0.0002)
Variance in skill gap	0.0012 (0.0006)	0.0038*** (0.0007)	0.0067** (0.0020)	0.0025*** (0.0005)
Win rate	−0.0030*** (0.0002)	−0.0035*** (0.0001)	−0.0022*** (0.0004)	−0.0038*** (0.0001)
Winning streak rate	−0.0090*** (0.0003)	−0.0086*** (0.0003)	−0.0113*** (0.0007)	−0.0101*** (0.0002)
Losing streak rate	0.0002 (0.0003)	0.0010** (0.0004)	−0.0014 (0.0010)	0.0002 (0.0002)
Controls				
Play count	−0.0539*** (0.0011)	−0.0907*** (0.0021)	−0.0615*** (0.0032)	−0.0708*** (0.0009)
Paid amount	0.0005 (0.0007)	0.0008 (0.0009)	0.0028 (0.0032)	−0.0001 (0.0005)
Cash earned	−0.0953*** (0.0025)	−0.0759*** (0.0019)	−0.1917*** (0.0070)	−0.0799*** (0.0016)
Cash spent	−0.0202*** (0.0019)	−0.0229*** (0.0016)	0.0248*** (0.0047)	−0.0302*** (0.0013)
Cash remained	−0.0146*** (0.0038)	0.0590*** (0.0031)	0.2192*** (0.0091)	−0.0132*** (0.0025)
Gold earned	−0.0274*** (0.0020)	−0.0082*** (0.0013)	−0.0266*** (0.0042)	−0.0139*** (0.0011)
Gold spent	−0.0159*** (0.0016)	−0.0109*** (0.0012)	0.0232*** (0.0031)	−0.0221*** (0.0010)
Gold remained	0.0806*** (0.0050)	0.0845*** (0.0044)	0.2664*** (0.0111)	0.0482*** (0.0034)
Players	120,066	142,632	28,094	234,604
Observations	891,793	695,944	114,427	1,473,310
Pseudo- R^2	.427	.210	.465	.265
BIC	230,869	320,284	37,837	542,014

Note. *** $p < .001$, ** $p < .01$, * $p < .05$; the standard errors are in parentheses.

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